

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

CASE STUDY

Group No : C15

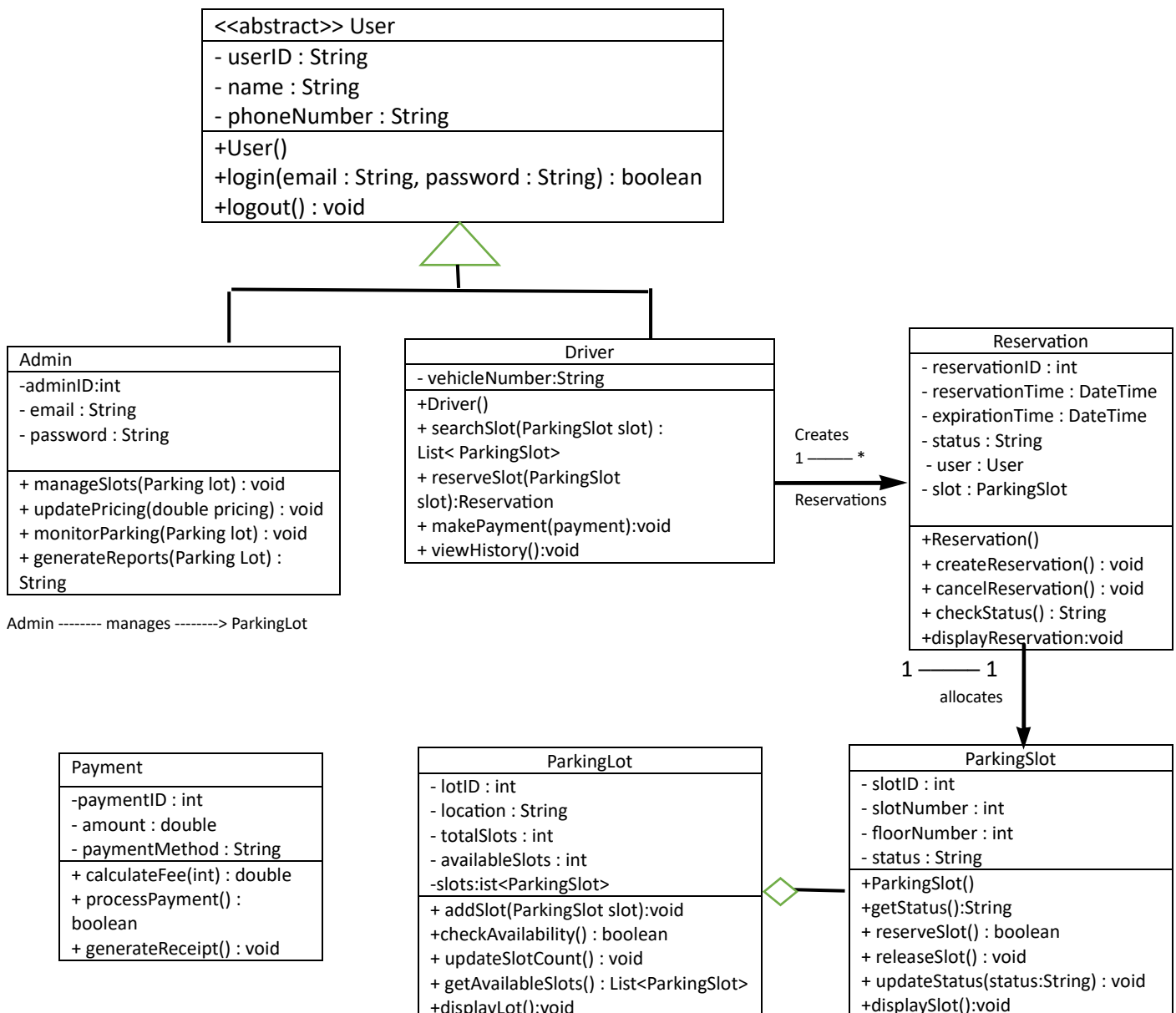
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1. Problem Title: Smart Parking Management System

STRUCTURAL DESIGN

2. Class diagram



Reservation ----- makes -----> Payment
1 — 1

BEHAVIUIORAL DESIGN

3. Use-case diagram

a. List of actors

1. Driver
2. Admin

b. List of use-cases

1. Search Parking Slots

-Allows the user to search for parking slots based on location and availability.

2. Check Availability

-Displays the current availability of parking slots in the parking lot.

3. Reserve Parking Slot

-Enables the user to select and reserve a parking slot in advance.

4. Create Reservation

-Creates a reservation record for the selected parking slot with time details.

5. Cancel Reservation

-Allows the user to cancel an existing reservation before the expiry time.

6. Check Reservation Status

-Shows whether a reservation is active, expired, or cancelled.

7. Make Payment

-Allows the user to pay the parking fee using digital payment methods.

8. Calculate Parking Fee

-Calculates the parking fee based on the duration of parking.

9. Process Payment

-Handles the transaction process and confirms successful payment.

10. Generate Receipt

-Generates a digital receipt after successful payment.

11. View Booking History

- Allows the user to view past and current parking bookings.

12. Manage Parking Slots

- Allows the admin to add, update, or remove parking slots.

13. Update Pricing

- Enables the admin to modify parking charges based on time or demand.

14. Monitor Parking

- Allows the admin to monitor slot occupancy and availability in real-time.

15. Generate Reports

- Generates analytical reports such as peak usage and occupancy trends.

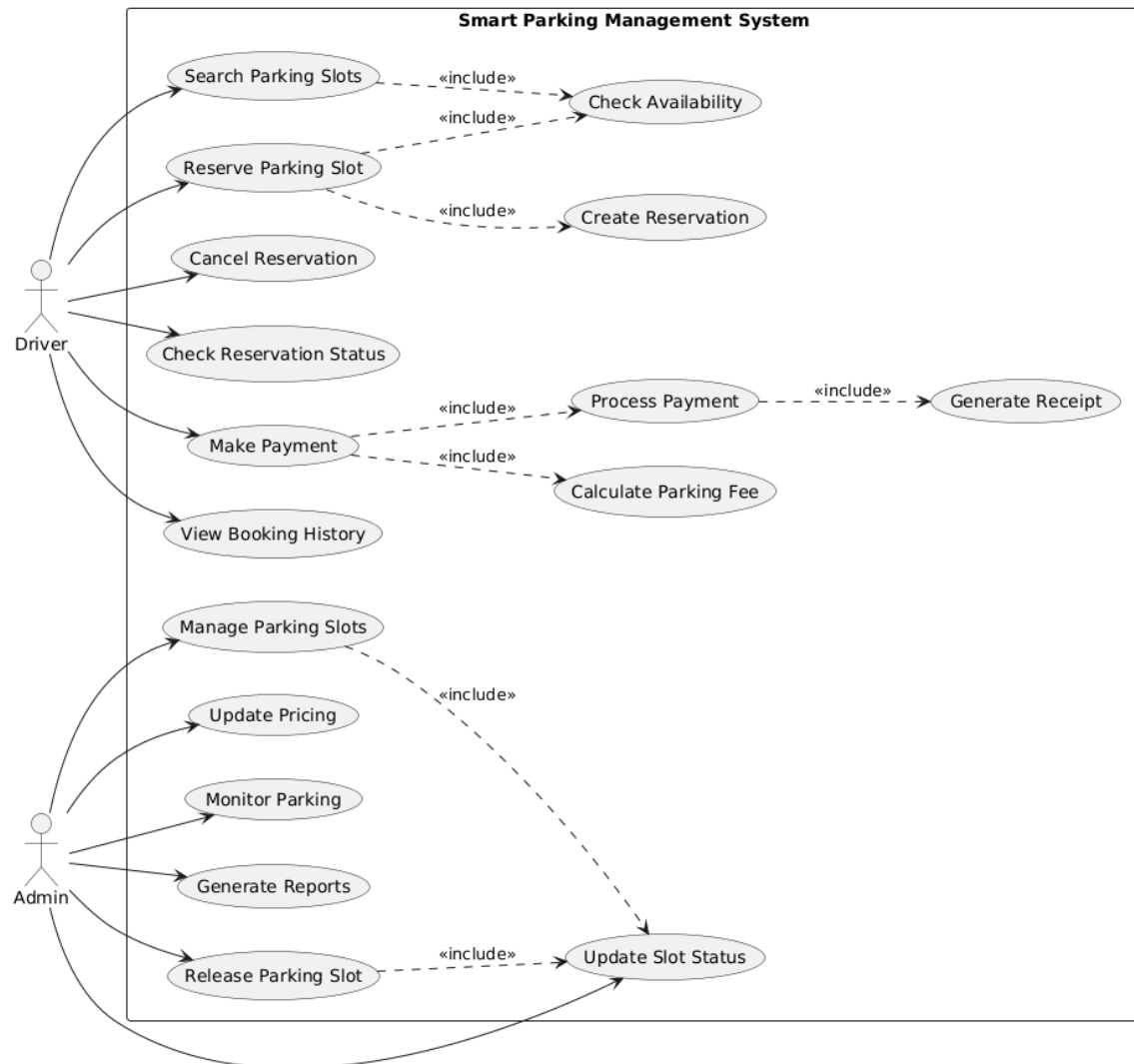
16. Update Slot Status

- Updates the status of a parking slot (occupied/free).

17. Release Parking Slot

- Frees up a parking slot when a vehicle exits.

c. Use case diagram



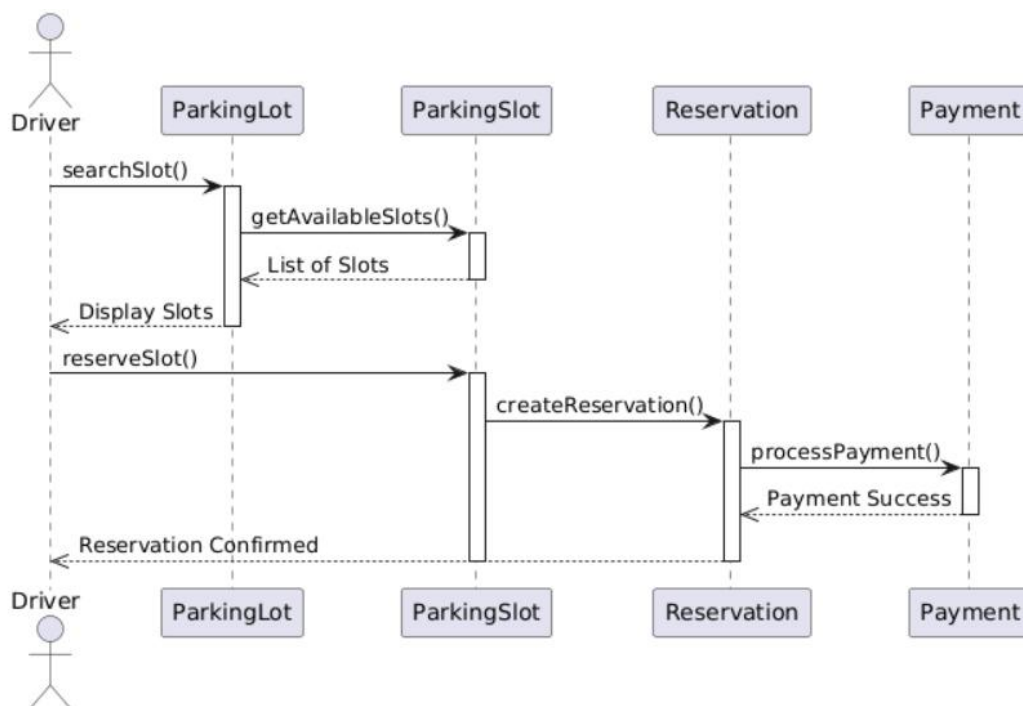
4. Sequence diagram

Use Case 1: Slot Reservation Process

Scenario Description:

A driver searches for an available parking slot in the parking lot. The system checks for available slots and displays them. The driver selects a slot and creates a reservation. After successful reservation, payment is processed and confirmation is generated.

UML Sequence Diagram:

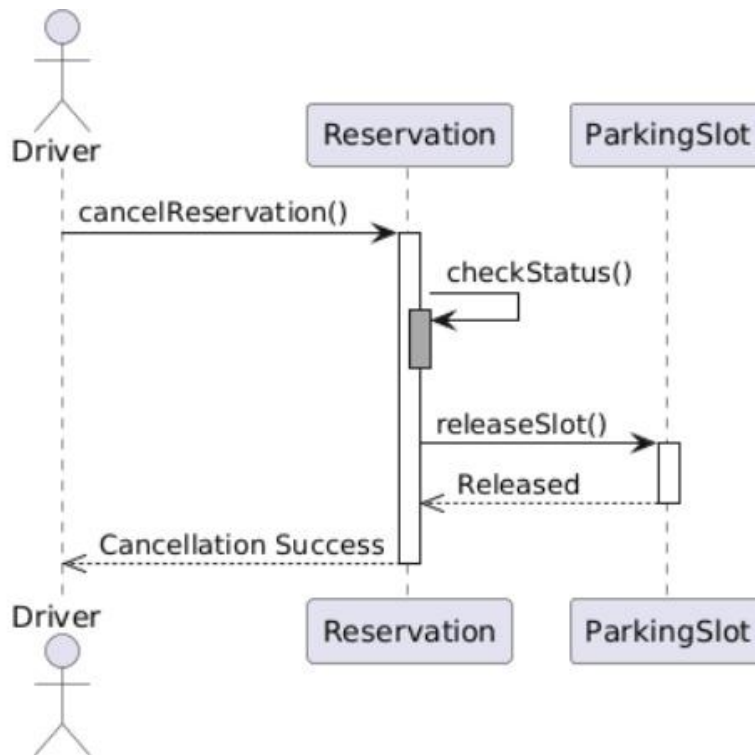


Use Case 2: Cancel Reservation Process

Scenario Description:

A driver cancels an already booked parking reservation. The reservation status is updated, and the slot becomes available again.

UML Sequence Diagram:



Use Case 3: Admin Managing Parking Slots

Scenario Description:

The admin monitors parking slots and updates slot pricing or availability. The system reflects updated parking information.

UML Sequence Diagram:

